Physics

Class 10 (English Medium)

Solved Exercises — Sindh Textbook Board

11 lessons — 327 questions

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Question: Self Assessment Q1: Find the frequency of a second's pendulum.

Answer: A second's pendulum has a period of 2 s, so its frequency = 1 / 2 = 0.5 Hz.

Question: Self Assessment Q2: Why is the weight of a simple pendulum's bob the cause of its simple harmonic motion?

Answer: The component of the bob's weight acting along the arc pushes it back toward the mean position. That component acts as the restoring force, so the weight itself makes the motion simple harmonic.

Question: Self Assessment Q3: At which point of a simple pendulum is the acceleration greatest, and what is it there?

Answer: The acceleration is greatest at the extreme points, because there the restoring force is largest and directed toward the mean position. At the mean position the acceleration is zero.

Question: Self Assessment Q4: In the bowl, the normal force acts opposite to the weight — so why does the ball not keep going up?

Answer: The normal force only balances the component of the weight that is perpendicular to the bowl's surface. It has no component along the slope, so it cannot lift the ball. The component of the weight along the slope keeps pulling the ball back down toward the bottom.

Question: Self Assessment Q5: At which points is the ball in the bowl slow, and at which point is it fast?

Answer: The ball comes momentarily to rest at the two extreme points (slowest), and is fastest at the bottom (mean position).

Question: Self Assessment Q6: If the vibration of a drum skin were not damped, what would happen?

Answer: The vibration would continue with the same amplitude and the sound would continue at the same loudness — the sound would not gradually fade away.

Question: 1. A stone is thrown into a pond and the waves travel to the shore. An observer sees 10 waves strike the shore in 50 seconds. What is the frequency of these waves? (a) 0.5 Hz (b) 15 Hz (c) 2.0 Hz (d) 50 Hz

Answer: 0.2 Hz.

Working: frequency = 10 / 50 = 0.2 Hz.

[Note: none of the printed options matches this correct value — the printed question/options appear to be misprinted.]

2. Water waves can show reflection, refraction and diffraction, all three. Which row of the table below is correct?

✓ (a) Reflection: yes, Refraction: yes, Diffraction: yes

(b) Reflection: yes, Refraction: no, Diffraction: yes

(c) Reflection: no, Refraction: yes, Diffraction: no

(d) Reflection: no, Refraction: no, Diffraction: no

3. Which of the four ripple-tank diagrams below correctly shows the refraction of water waves entering shallow water from deep water?

(a) A

(b) B

✓ (c) C

(d) D

4. In the diagram a water wave approaches a gap in a barrier, and point P lies behind the barrier. What effect acts on the wave at P?

✓ (a) Diffraction

(b) Spreading out

(c) Reflection

(d) Refraction

5. Water waves bend as they go from deep to shallow water. Which of their properties stays the same?

(a) Direction

✓ (b) Frequency

(c) Speed

(d) Wavelength

6. _____ is not a property of a wave.

(a) Amplitude

(b) Period

✓ (c) Mayu

(d) Speed

7. During SHM, at which point does the moving body have its greatest speed?

(a) Affected point

(b) Lower point

✓ (c) Equilibrium point

(d) End point

8. The acceleration of a vibrating pendulum bob at the extreme limit is _____.

✓ (a) Achalta

(b) Chak

(c) Air

(d) Gravity

9. In the ball-and-bowl system the central point is _____.

(a) The ground

✓ (b) The floor of the bowl

(c) The centre of the bowl

(d) The extreme limit

10. Vibrational motion is kept going by _____.

(a) Straight motion

(b) Returning force

(c) Joining force

(d) Mechanical force

Question: Section (B) Structured Question 1: Define a transverse wave.

Answer: A transverse wave is a wave in which the particles of the medium vibrate at right angles to the direction in which the wave travels. It is made up of crests and troughs. Examples: water waves, light waves.

Question: Section (B) 2: Define a longitudinal wave.

Answer: A longitudinal wave is a wave in which the particles of the medium vibrate along (in the same direction as) the direction in which the wave travels. It is made up of compressions and rarefactions. Examples: sound waves, slinky waves.

Question: Section (B) 3: (a) Write a short note on mechanical waves. (b) Why do mechanical waves need a medium?

Answer: (a) Mechanical waves are waves that travel in a material medium. They carry energy from one place to another by the vibration of the medium's particles. Examples: sound, water waves, slinky waves.

(b) Because they are produced by the vibration of the medium's particles and move on by passing the disturbance from one particle to the next. Where there are no particles at all (a vacuum), there is nothing to vibrate, so they cannot travel.

Question: Section (B) 4: Give everyday examples of energy transfer without transfer of matter.

Answer: (1) A cork floating on water: as a wave passes, the cork bobs up and down but does not travel along with the wave.

(2) The sound of a drum: the sound reaches the ear, but the air does not flow from the drum to the ear.

(3) A stadium wave: people stand and sit in their own places without moving away — the wave travels around the stadium.

Question: Section (B) 5: (a) Define these properties of waves: (1) amplitude (2) period (3) frequency (4) wavelength. (b) Derive the equation $V = f\lambda$.

Answer: (a) (1) Amplitude: the maximum displacement of a particle from the mean position.

(2) Period: the time needed for one complete vibration.

(3) Frequency: the number of vibrations in one second.

(4) Wavelength: the distance between two nearest identical points (such as two crests or two compressions).

(b) Speed = distance / time.

In one period a wave covers one wavelength, so $V = \lambda / T$.

And frequency $f = 1 / T$, so $V = f\lambda$.

Question: Section (B) 6: (a) What is a ripple tank? (b) Define a wave front.

Answer: (a) A ripple tank is a shallow tray of water with a lamp above it

and a white screen below. A vibrating bar makes waves in the water, and their shadows on the screen show the wave fronts.

(b) A wave front is the line joining the points of a wave that are in the same phase (for example, all the crests).

Question: Section (B) 7: Describe the ripple-tank experiment for the reflection of waves.

Answer: Straight waves strike a metal barrier and return into the water. The angle of incidence equals the angle of reflection. The speed and the wavelength of the waves stay the same.

Question: Section (B) 8: Describe the refraction / diffraction of waves.

Answer: Refraction: when waves enter shallow water from deep water they slow down, the wavelength shortens, and the direction bends toward the normal (if the waves arrive at an angle). Diffraction: when waves pass through a narrow gap or past the edge of an obstacle they spread out and reach into the shadow region. The spreading is greater when the gap is small compared with the wavelength.

Question: Section (B) 9: Simple Harmonic Motion (SHM): (a) What is SHM? (b) What are the necessary conditions for SHM?

Answer: (a) SHM is motion in which the acceleration is proportional to the displacement from the mean position and is always directed toward the mean position.

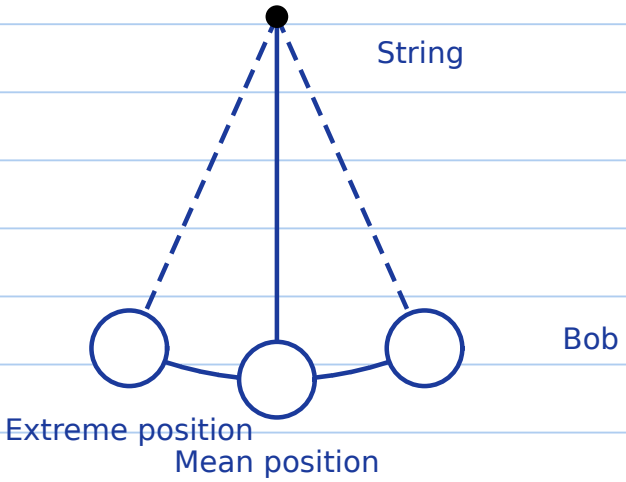
(b) Necessary conditions: (1) a restoring force proportional to the displacement; (2) that force always directed toward the mean position; (3) inertia in the body, so that it overshoots the mean position and the vibration continues.

Question: Section (B) 10: (a) Describe the SHM of a simple pendulum with

a diagram. (b) In a simple pendulum the time period is T = 2π√(L/g). State its period.

Answer: (a) When a bob hanging from a string is pulled to one side and released, the component of its weight along the arc pushes it back toward the mean position. The bob overshoots the mean position, climbs the other side, and returns — for small angles this vibration is approximately SHM.

(b) The period depends only on the length of the string and the acceleration due to gravity: a longer string gives a longer period, and it does not depend on the mass of the bob or on the (small) amplitude.



Labelled diagram — schematic; parts as named in the answer. See the book diagram for exact form.

Question: Section (B) 11: (a) Describe SHM with the help of the ball-and-bowl system. (b) At which point is the ball's speed greatest during SHM in the bowl?

Answer: (a) A ball released on one side of a smooth bowl rolls down, overshoots the mean position, climbs the other side, and returns. This to-and-fro motion is approximately SHM.

(b) The speed is greatest at the bottom (mean position).

Question: Section (B) 12: (a) What are damped oscillations? (b) How does damping reduce the amplitude of the vibration of a wave? (c) Why does the amplitude decrease with each swing?

Answer: (a) Damped oscillations are oscillations whose amplitude gradually decreases because energy is lost (for example to friction or air resistance).

(b) The resisting force acts against the motion and removes a little energy in each swing, so each swing reaches a smaller height than the one before.

(c) Because in every swing a part of the energy is spent against the resisting force, so less energy remains for the next swing.

Question: Section (C) Numerical Question 1: A radio station broadcasts at a frequency of 1300 kHz. Radio waves travel in space at 3x10^8 m/s. Find the wavelength of these radio waves.

Answer: λ = speed / frequency = 3x10^8 / 1.3x10^6 = 230.76 m.

Question: Section (C) 2: The wavelength of water waves is 1.6 m and the frequency is 0.80 Hz. Find the speed of the water waves.

Answer: Speed = frequency x wavelength = 0.80 x 1.6 = 1.28 m/s.

Question: Section (C) 3: 50 waves pass a point in 10 seconds and their wavelength is 8 cm. Find their frequency, period and speed.

Answer: Frequency = 50 / 10 = 5 Hz.

Period = 1 / 5 = 0.2 s.

Speed = 5 x 0.08 = 0.40 m/s.

Question: Section (C) 4: Waves produced in a Slinky travel at 0.40 metres per second and their frequency is 20 hertz. What is the least distance between two consecutive compressions?

Answer: The distance between two consecutive compressions = wavelength = speed / frequency = 0.40 / 20 = 0.02 m.

[Note: the printed speed figure reads 40; the printed answer 0.02 m fits 0.40 m/s, so the decimal point is missing in the scan.]

Question: Section (C) 5: Suppose a wave on a Slinky completes one full vibration in 0.40 s and the wavelength is 0.60 m. Find the wave's (a) period and frequency (b) speed.

Answer: (a) Period = 0.40 s;

frequency = 1 / 0.40 = 2.5 Hz.

(b) Speed = wavelength / period = 0.60 / 0.40 = 1.5 m/s.

Question: Section (C) 6: If 80 compressions of a wave pass a point of a spring in 20 seconds and the distance between consecutive compressions is 8 cm, find the frequency, period and speed of the wave.

Answer: Frequency = 80 / 20 = 4 Hz.

Period = 1 / 4 = 0.25 s.

Speed = 4 x 0.08 = 0.32 m/s.

Question: Section (C) 7: In a pond, waves spread from one shore at 0.9 metres per second. They travel to the other shore and return in 30.0 seconds. Find the total distance travelled by the wave and the distance to the other shore.

Answer: Total distance = 0.9 x 30.0 = 27 m.

Distance to the other shore = 27 / 2 = 13.5 m.

Question: Section (C) 8: The string of a simple pendulum is 80.0 cm long. Find its (a) period (b) frequency when the acceleration due to gravity is 9.8 metres per second squared.

Answer: L = 0.80 m.

(a) $T = 2\pi\sqrt{(L/g)} = 2\pi\sqrt{(0.80/9.8)} = 1.794$ s.

(b) f = 1 / 1.794 = 0.557 Hz.

آواز11یونٹ :

Question: Self Assessment Q1: Which is faster, light or sound?

Answer: Light is faster — about 3x10^8 m/s in a vacuum, while sound in air travels at only about 340 m/s.

Question: Self Assessment Q2: Is it possible to produce a distinct echo in a 10-metre room?

Answer: No. To hear a distinct echo the sound must return after at least 0.1 s, which needs the reflecting surface to be at least 17 m away. In a 10 m room the echo merges with the original sound.

1. Sound is a form of _____ energy.

(a) Electrical

✓ (b) Mechanical

(c) Heat

(d) Chemical

2. The hearing frequency range of a normal human ear is _____.

(a) 10 Hz to 10 kHz

✓ (b) 20 Hz to 20 kHz

(c) 25 Hz to 25 kHz

(d) 30 Hz to 30 kHz

3. The speed of sound at 0°C is _____.

✓ (a) 331 metres per second

(b) 34 metres per second

(c) 17 metres per second

(d) 680 metres per second

4. Sound travels fastest through a solid because _____.

(a) Gas molecules are far apart

(b) Sound does not travel fast in a gas

✓ (c) The molecules of a solid are close together

(d) Gas molecules are fast with motion

5. Which two factors affect the speed of sound in air?

(a) The amount of air

(b) The measure of heat in air

(c) The measure of air and the body

✓ (d) Temperature and humidity in air

6. The distance between two consecutive compressions is called the _____.

(a) Time period

(b) Amplitude

(c) Frequency

✓ (d) Wavelength

7. State the order of the media below from fastest to slowest speed of sound:

(a) Gas, water, iron

(b) Water, iron, gas

✓ (c) Iron, water, gas

(d) Gas, iron, water

8. What are the uses of ultrasound in industry?

(a) Absorption

✓ (b) Scanning inside the body

(c) Catching

(d) Measuring air humidity

9. An echo is caused by what?

(a) Absorption

(b) Being caught

(c) Reflection

(d) Refraction

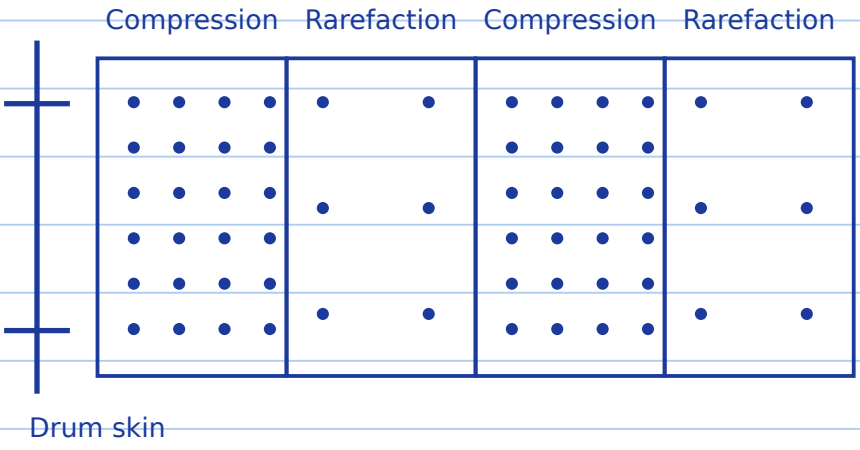
Question: 10. Which of the following waves cannot travel in a vacuum? (a) Sound wave (b) Infrared radiation (c) Mechanical wave (d) X-ray

Answer: (a) Sound wave. [Note: sound is a mechanical wave, so (c) is also true as a general statement; the book's intended answer is (a).]

Question: Section (B) 1: (a) How is sound produced? (b) How are compressions and rarefactions of sound produced in air? Explain with a diagram.

Answer: (a) Sound is produced by a vibrating body.

(b) The vibrating body pushes the layers of air together in front of it (compression), and as it moves back they spread apart (rarefaction). These compressions and rarefactions travel onward through the air one after another — that is the sound wave. (Diagram: the vibrating drum skin and the bands of compression/rarefaction in the air.)



Labelled diagram — schematic; parts as named in the answer. See the book diagram for exact form.

Question: Section (B) 2: (a) Why is a sound wave called a mechanical wave? (b) Why does sound need a medium to travel? Describe with

reference to an experiment.

Answer: (a) Because sound travels by the vibration of the particles of a medium — that is the definition of a mechanical wave.

(b) Without particles the vibration cannot move on. Evidence: the bell-jar experiment — as the air is pumped out of a glass jar the bell's sound fades away, and when air is let back in the sound returns.

Question: Section (B) 3: (a) State the difference between musical sound and noise. (b) State why noise is harmful to humans.

Answer: (a) Musical sound is regular, pleasant, periodic vibration; noise is irregular, unpleasant, non-periodic vibration.

(b) Loud and prolonged noise damages hearing, disturbs sleep and concentration, and raises stress and blood pressure.

Question: Section (B) 4: (a) Define the pitch or quality of sound. (b) Is it possible to combine the waves of the sounds of two different musical instruments into one wave?

Answer: (a) Pitch is the highness or lowness of a sound, fixed by frequency (higher frequency = higher pitch). Quality (timbre) is the character by which two sounds of the same pitch and loudness from different sources can be told apart.

(b) Yes — waves superpose and combine into a resultant wave; the ear can still tell them apart by their difference in quality.

Question: Section (B) 5: (a) What is the speed of sound in three media: gas, water and iron? (b) State the effect on the speed of sound in air of: (1) temperature (2) humidity.

Answer: (a) Fastest in iron, then in water, slowest in a gas.

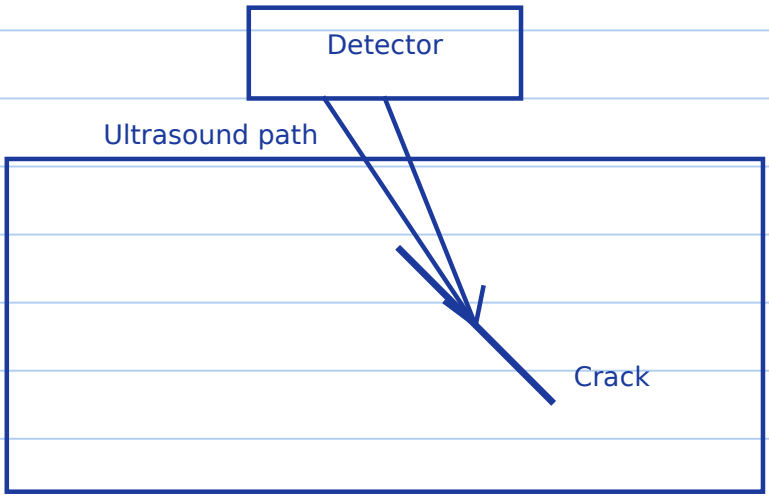
(b) (1) Temperature: speed rises as temperature rises (about 0.6 m/s per 1°C). (2) Humidity: speed is greater in moist air than in dry air.

Question: Section (B) 6: (a) Define an echo. (b) State the uses of Sonar. (c) How can ultrasound locate a crack in a metal block? Explain with a diagram.

Answer: (a) An echo is a sound heard again when it returns after reflection from a distant surface.

(b) Sonar is used to find the depth of the sea, shoals of fish, submarines and underwater obstacles.

(c) Ultrasound is sent into the block; it reflects from the crack and the signal differs from that of a sound block, which locates the crack.
(Diagram: the ultrasound path, reflection from the crack, detector.)



Labelled diagram — schematic; parts as named in the answer. See the book diagram for exact form.

Question: Section (B) 7: (a) Define: (i) infrasound (ii) the audible frequency range (iii) ultrasound. (b) Why is ultrasound used for cleaning? (c) State the uses of ultrasound in medical diagnosis and treatment in hospitals.

Answer: (a) (i) Infrasound: sound of frequency below 20 Hz. (ii) Audible

range: 20 Hz to 20 kHz. (iii) Ultrasound: sound of frequency above 20 kHz.

(b) Its rapid vibration shakes dirt and grease off small and delicate objects in a cleaning liquid, reaching places a brush cannot.

(c) Scanning organs and the baby in the womb (ultrasonography), echocardiography of the heart, breaking kidney stones, physiotherapy.

Question: Section (C) 1: Calculate the speed of sound in air at 50°C. The speed of sound at 0°C is 331 metres per second.

Answer: Speed = 331 + 0.6 x 50 = 331 + 30 = 361, about 360 m/s.

(The book prints 360.0 m/s.)

Question: Section (C) 2: A human's hearing range is from 20 Hz to 20 kHz. Calculate the difference in wavelength of these sounds. The speed of sound is 340 metres per second.

Answer: Longest wavelength = 340 / 20 = 17 m.

Shortest wavelength = 340 / 20000 = 0.017 m.

[Note: the book prints 17.2 m, which fits 344 m/s; with the printed 340 m/s the value is 17 m.]

Question: Section (C) 3: A ship finds the depth of the sea beneath it using ultrasound waves. Its echo returns after 40 milliseconds. The speed of sound in water is 1480 metres per second. Find the depth of the sea.

Answer: Time = 0.040 s.

Depth = speed x time / 2 = 1480 x 0.040 / 2 = 29.6 m, about 30 m.

Question: Section (C) 4: At night a bat emits sound waves to hunt. The

wavelength of this sound is 0.0080 m. The speed of sound is 340 metres per second. (i) Calculate the frequency of the sound. (ii) The sound wave strikes the prey and the returning wave is heard after 0.10 seconds. How far did the wave travel? (iii) How far away was that distance from the prey?

Answer: (i) Frequency = 340 / 0.0080 = 42500 Hz = 42.5 kHz.

(ii) Total distance = 340 x 0.10 = 34 m.

(iii) Distance to the prey = 34 / 2 = 17 m.

برق مقناطيسي پتي12يونټ :

Question: Self Assessment Q1: State the health concerns linked to the energy component of electromagnetic radiation.

Answer: Higher-energy rays (UV, X, gamma) damage cells and tissues — skin burns, eye damage, and cancer from long or intense exposure.

Question: Self Assessment Q2: What are the advantages of optical fibres in communications?

Answer: They carry very many signals, the loss is very low, they are light and thin, there is no electrical interference, and signals travel fast and safely.

Question: Self Assessment Q3: What role do gamma rays play in radiosurgery?

Answer: Many weak gamma rays are focused on the tumour; together they destroy it without cutting the body (gamma knife).

1. The waves that carry great energy and are used to treat cancer cells are:

(a) Ultraviolet rays

(b) Microwaves

✓ (c) Gamma rays

(d) Radio waves

2. The speed of light in diamond is _____.

✓ (a) 1.2x10^8 metres per second

(b) 5x10^8 metres per second

(c) 1.2x10^10 metres per second

(d) 2.5x10^8 metres per second

3. Electromagnetic rays are used in radiotherapy to destroy cancer cells.

They are _____.

(a) Infrared rays

(b) Visible light

✓ (c) Gamma rays

(d) Ultraviolet rays

4. The group that consists of electromagnetic waves is:

✓ (a) Light waves, radio waves, microwaves

(b) Light waves, radio waves, sound waves

(c) Light waves, sound waves, microwaves

(d) Radio waves, sound waves, microwaves

5. The list that shows electromagnetic waves in order of decreasing wavelength:

✓ (a) Microwaves, X-rays, gamma rays

(b) Microwaves, gamma rays, X-rays

(c) X-rays, gamma rays, microwaves

(d) Gamma rays, X-rays, microwaves

6. The type of electromagnetic wave used in a night-time security scanner is _____.

✓ (a) Infrared

(b) Microwaves

(c) Radio waves

(d) X-rays

7. When a thin ray of light enters a glass prism from air, which property of the wave does not change?

(a) Direction

✓ (b) Frequency

(c) Speed

(d) Wavelength

8. In a television remote control, _____ type of waves are used.

(a) Radio waves

✓ (b) Infrared waves

(c) Ultraviolet waves

(d) Visible light

9. The _____ colour bends the most in a prism.

✓ (a) Violet

(b) Green

(c) Dark

(d) Yellow

10. The phenomenon in which white light splits into seven colours is called _____.

(a) Reflection

(b) Refraction

✓ (c) Dispersion

(d) Diffraction

Question: Section (B) 1: (a) Define the dispersion of light. (b) Describe the dispersion of light when it passes through a prism.

Answer: (a) Dispersion of light is the splitting of white light into its seven colours.

(b) White light entering a prism bends and splits into seven colours: violet, indigo, blue, green, yellow, orange, red. Violet bends the most, red the least.

Question: Section (B) 2: (a) Explain how a rainbow is formed. (b) Explain how each colour is linked to a particular frequency / wavelength.

Answer: (a) After rain, the water drops in the air act like tiny prisms. Sunlight disperses in them, and the observer's eye sees the seven colours

as an arc.

(b) Each colour has its own frequency and wavelength: red has the longest wavelength and lowest frequency, violet the shortest wavelength and highest frequency.

Question: Section (B) 3: (a) What are electromagnetic waves? (b) List in order: the electromagnetic spectrum in decreasing wavelength. Comment on the statement that ultraviolet rays have a higher frequency than radio waves because UV can travel faster in a vacuum.

Answer: (a) Electromagnetic waves are transverse waves of vibrating electric and magnetic fields that travel in a vacuum at 3x10^8 m/s with no medium.

(b) Decreasing wavelength: radio waves, microwaves, infrared, visible light, ultraviolet, X-rays, gamma rays. Comment: the UV frequency is indeed higher than radio, but the reason is not speed — all electromagnetic waves travel at the same speed in a vacuum; the frequency is higher because the wavelength is shorter.

Question: Section (B) 4: Present the properties of ultraviolet rays and radio signals. (a) Which of them travels faster? (b) Which of them has the higher frequency? (c) Which of them has the longer wavelength?

Answer: (a) Both travel at the same speed, 3x10^8 m/s, in a vacuum.

(b) Ultraviolet has the higher frequency.

(c) Radio waves have the longer wavelength.

Question: Section (B) 5: (a) What are the sources of radio waves? (b) What are the advantages of using radio waves in communication?

Answer: (a) Sources: transmitting antennas (radio/TV stations), and natural sources such as stars.

(b) Advantages: they cover long distances, pass around buildings and obstacles, and carry radio, TV and mobile-phone signals.

Question: Section (B) 6: What is the importance of microwaves in satellite communication?

Answer: Microwaves pass through the atmosphere, they can be aimed at a satellite in a narrow beam, and they carry large amounts of information (TV, phone, internet).

Question: Section (B) 7: (a) In what way do molecules emit infrared radiation? (b) How do security staff take thermal images at night?

Answer: (a) All warm bodies emit infrared because their molecules move/vibrate; the hotter the body, the more infrared.

(b) An infrared camera picks up the heat (infrared) given off by bodies and shows the temperature differences as an image, so people and objects are seen even in the dark.

Question: Section (B) 8: (a) Give examples of optical fibres in daily life: (i) communication (ii) medicine and industry. (b) On which principle do optical fibres work?

Answer: (a) (i) Communication: telephone, internet and cable TV signals are sent through fibres. (ii) Medicine and industry: an endoscope is used to see inside the body.

(b) They work on the principle of total internal reflection.

Question: Section (B) 9: (a) How can exposure to sunlight damage the skin? (b) Why are sunbeds used under supervision?

Answer: (a) The UV in sunlight burns the skin, and long exposure can damage cells and cause cancer.

(b) Because their UV can also burn and damage the skin, so the time and dose must be kept under control.

Question: Section (B) 10: (a) Explain a luminous body. (b) Describe how light spreads from a luminous body.

Answer: (a) A luminous body is one that gives out its own light, such as the Sun, a bulb, a candle.

(b) Light spreads from it in straight lines in all directions. [Note: the printed wording of (b) is unclear.]

Question: Section (B) 11: Explain how X-rays are used to find cracks in iron.

Answer: X-rays pass through the metal. Less is absorbed at a crack, so on the film/plate the crack appears as a dark line and the flaw is revealed.

Question: Section (B) 12: (a) What are gamma rays? (b) How are gamma rays used in radiosurgery to destroy cancer cells? (c) Explain how gamma rays are used for medical imaging in hospitals.

Answer: (a) Gamma rays are the highest-energy, shortest-wavelength electromagnetic waves, given out by radioactive nuclei.

(b) Many weak rays are focused on the tumour so that the tumour receives a strong dose while the surrounding healthy tissue receives little.

(c) In a PET scan, gamma rays from a tracer are caught by detectors and built into a 3-D image of the organ's activity.

Question: Section (C) 1: An electromagnetic wave has a wavelength of

0.15µm, said to be an infrared ray in the classification of the electric spectrum. What is its frequency? The speed of light is 3x10^8 m/s.

Answer: λ = 0.15 µm = 1.5x10^-7 m.

Frequency = 3x10^8 / 1.5x10^-7 = 2x10^15 Hz.

This is ultraviolet (150 nm), not infrared.

[Note: the book prints 2x10^12 Hz, which fits λ = 0.15 mm (infrared);

the printed unit/answer are inconsistent.]

Question: Section (C) 2: An ultraviolet ray is used in eye laser surgery. Its wavelength is 15.0 nm — what is its frequency?

Answer: λ = 15.0 nm = 1.5x10^-8 m.

Frequency = 3x10^8 / 1.5x10^-8 = 2.0x10^16 Hz.

[Note: the book prints 1.55x10^15 Hz, which fits λ = 193 nm (excimer laser);

with the printed 15.0 nm the value is 2.0x10^16 Hz.]

Question: Section (C) 3: In an MRT unit the frequency of the radio wave used is 100 MHz. Calculate its wavelength.

Answer: λ = 3x10^8 / 1x10^8 = 3 m.

Question: Section (C) 4: The distance from the Earth to the Sun is 1.49x10^11 m. How long does a radio wave leaving the Sun take to reach the Earth?

Answer: Time = distance / speed = 1.49x10^11 / 3x10^8 = 496.67 s

(about 8.3 minutes).

Question: Section (C) 5: Distances in space are measured in light years — that is, the distance light travels in one year. Find that distance in kilometres.

Answer: One year = 365 x 24 x 3600 = 31,536,000 s.

Distance = 3x10^8 x 31,536,000 = 9.46x10^15 m = 9.46x10^12 km.

[Note: the book prints 9.3x10^12 km.]

Question: Section (C) 6: Green light has a wavelength of (5.5x10^-7 m). What is its frequency?

Answer: Frequency = 3x10^8 / 5.5x10^-7 = 5.45x10^14 Hz.

[Note: the book prints the wavelength as 5.5x10^7 m — the minus sign is missing.]

Question: Section (C) 7: An ordinary household microwave oven has a frequency of 2.45 GHz — what is the wavelength of its rays?

Answer: λ = 3x10^8 / 2.45x10^9 = 0.1224 m = 122.4 mm.

بصرياتي اوزارن جي جوڙجڪ13يونٽ :

Question: Self Assessment Q1: Why do distant objects appear blurred in short sight?

Answer: Because the image of a distant object forms not on the retina but in front of it, so it looks blurred.

Question: Self Assessment Q2: What is the common treatment to correct the reflection defect of long sight?

Answer: A convex lens of suitable power — spectacles or a contact lens — which brings the image forward onto the retina.

1. In a plane mirror, the size of the image depends on _____.

✓ (a) The size of the object

(b) The place of the object

(c) The spread of the object

(d) The form of the object

2. In a normal human eye the image forms _____.

(a) In front of the retina

(b) Behind the retina

✓ (c) On the retina

(d) In the lens

3. When a ray of light enters a rarer medium from a denser one, it bends _____ the normal.

(a) Perpendicular to the normal

(b) Through the normal

(c) Toward the normal

✓ (d) Away from the normal

4. In a compound microscope the focal length of the eyepiece is _____.

(a) Zero

(b) Khato

(c) Small

✓ (d) Large

5. When the angle of refraction becomes 90° and the refractive index for water is 1.33, its critical angle becomes _____.

✓ (a) 48.8°

(b) 49.1°

(c) 50.0°

(d) 51.0°

6. To see stars in dim light we use _____.

(a) Compound microscope

(b) Simple microscope

(c) Endoscope

✓ (d) Telescope

7. The human eye works like _____.

✓ (a) A camera

(b) A projector

(c) A telescope

(d) A microscope

8. A magnifying glass does not produce an enlarged _____. (a) Real and inverted image (b) Real and erect image (c) Virtual and inverted image (d) Virtual and erect image

9. If true light is reflected back in the same medium, it is called total _____.

(a) Outside reflection

✓ (b) Internal reflection

(c) Outside refraction

(d) Inside refraction

10. The inner part of an optical fibre has a refractive index _____ than that of glass or plastic.

(a) Zero

✓ (b) Greater

(c) Smaller

(d) No refractive index

11. A magnifying glass is called _____.

(a) Endoscope

(b) Compound microscope

✓ (c) Simple microscope

(d) Telescope

12. The eye defect in which the image forms behind the retina is called _____.

✓ (a) Long sight

(b) Short sight

(c) Blind spot

(d) Image defect

13. Short sight is corrected with _____.

✓ (a) Concave lens

(b) Concave mirror

(c) Plane mirror

(d) Plane lens

14. A lens forms images because of _____.

(a) Dispersion

(b) Bending

✓ (c) Refraction

(d) Reflection

15. A dentist uses _____ to see the unreachable parts of the teeth.

(a) Plane mirror

✓ (b) Concave mirror

(c) Convex lens

(d) Plane lens

Question: Section (B) 1: (a) How do you understand the term reflection of light? (b) Explain reflection at a plane surface with a diagram. (c) Define these terms of reflection: (a) normal (b) incident ray (c) reflected ray (d) laws of reflection.

Answer: (a) Reflection of light is light striking a surface and returning into the same medium.

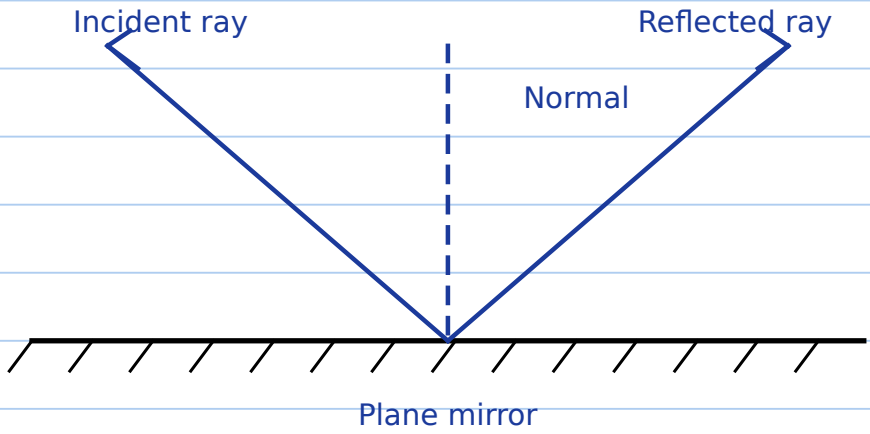
(b) At a plane surface the ray strikes and returns at the same angle; the incident ray, the reflected ray and the normal all lie in one plane.

(c) (a) Normal: the perpendicular to the surface at the point of incidence.

(b) Incident ray: the ray arriving at the surface.

(c) Reflected ray: the ray leaving the surface.

(d) Laws of reflection: (1) the angle of incidence equals the angle of reflection; (2) the incident ray, the reflected ray and the normal all lie in the same plane.



Labelled diagram — schematic; parts as named in the answer. See the book diagram for exact form.

Question: Section (B) 2: In each case below, which type of mirror is used? Give the reason: (a) A car's side / rear-view mirror (b) A searchlight / torch reflector (c) A dentist's mirror.

Answer: (a) Convex mirror — it gives a wide, erect, diminished image, so a large area behind is seen.

(b) Concave mirror — it sends the rays out as a parallel beam.

(c) Concave mirror — it gives an enlarged, erect image of the tooth.

Question: Section (B) 3: Refraction of light: (a) Define refraction of light. (b) Explain with a diagram the path of light through a glass slab. (c) Define these terms of refraction: (1) incident ray (2) refracted ray (3) laws of refraction.

Answer: (a) Refraction of light is the bending of light as it enters from one medium into another, because its speed changes.

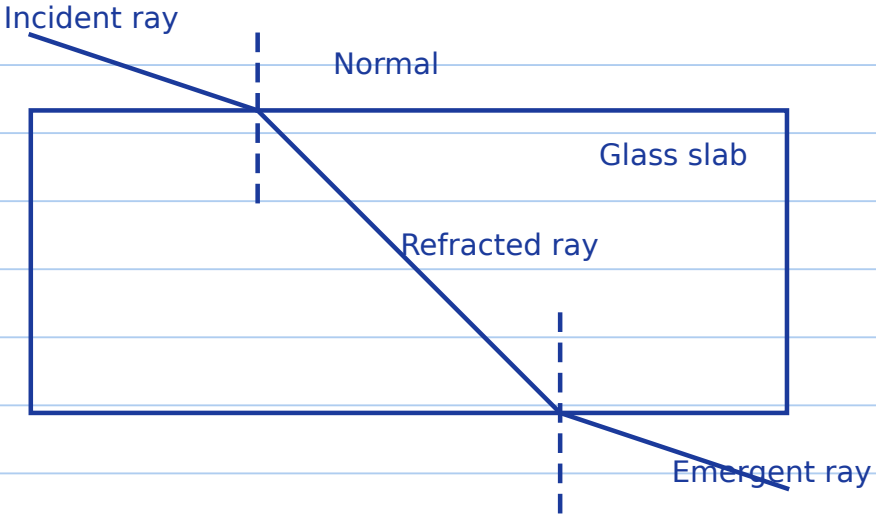
(b) In a glass slab the ray bends toward the normal on entering, travels straight inside, and bends away from the normal on leaving; the emerging ray is parallel to the incident ray (slightly shifted sideways).

(c) (1) Incident ray: the ray arriving at the surface.

(2) Refracted ray: the bent ray in the second medium.

(3) Laws of refraction (Snell's law): sin i / sin r = constant (refractive index);

the incident ray, the refracted ray and the normal lie in one plane.



Labelled diagram — schematic; parts as named in the answer. See the book diagram for exact form.

Question: Section (B) 4: (a) What is the refractive index of a medium? (b) Describe the experiment for the refractive index of a glass slab. (c) On what factors does the refraction of light depend?

Answer: (a) Refractive index n = speed of light in vacuum / speed of light in the medium; it tells how much the medium slows and bends light.

(b) Send a slanting ray onto the slab, mark the incident and emerging rays, measure i and r, then n = sin i / sin r.

(c) On the refractive indices of the two media, on the angle of incidence,

and on the colour (wavelength) of the light.

Question: Section (B) 5: (a) What is a prism? (b) Describe the passage of coloured light through a prism. (c) Suppose a ray of light enters a prism surface at (i) 0° to the normal (ii) 30° to the normal. What happens to the ray in each case — in terms of wavelength, frequency and speed?

Answer: (a) A prism is a block of transparent glass with two triangular ends and three rectangular faces.

(b) White light disperses in the prism into seven colours; violet bends the most, red the least.

(c) (i) At 0° the ray passes straight through without bending; direction, speed and wavelength do not change. (ii) At 30° the ray bends toward the normal on entering, slows down, the wavelength shortens, but the frequency stays the same.

Question: Section (B) 6: (a) What is a lens? (b) Describe the passage of light through a convex lens. (c) Describe in detail how a convex lens forms images.

Answer: (a) A lens is a piece of transparent glass/plastic with one or two curved surfaces that refracts light to form images.

(b) Rays parallel to the principal axis meet at the focal point; a ray through the centre goes straight.

(c) Object beyond 2F: real, inverted, diminished image between F and 2F. At 2F: real, inverted, same-size image at 2F. Between F and 2F: real, inverted, enlarged image beyond 2F. At F: image at infinity. Inside F: virtual, erect, enlarged image on the same side.

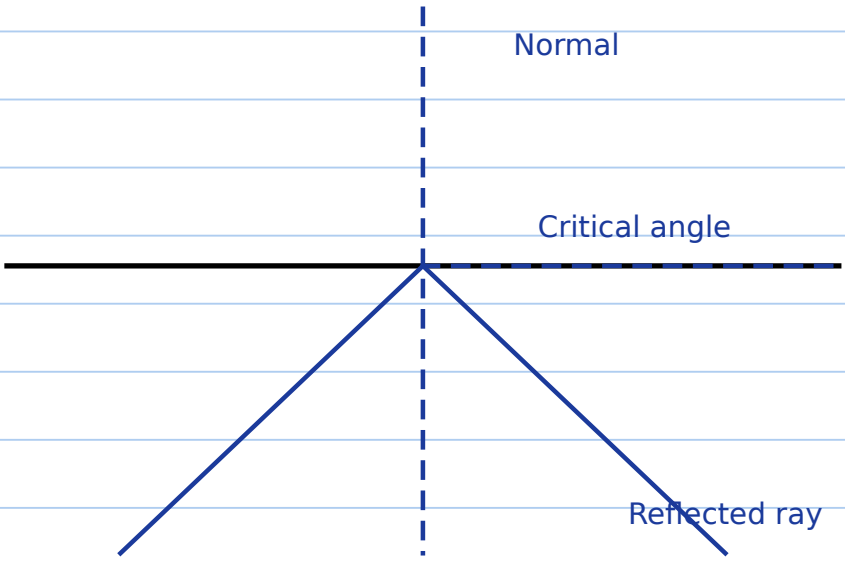
Question: Section (B) 7: Total internal reflection: (a) Define the critical angle. (b) Explain total internal reflection with a diagram. (c) State the conditions of total internal reflection. (d) Give everyday examples of total internal reflection.

Answer: (a) The critical angle is the angle of incidence in the denser medium for which the angle of refraction becomes 90°.

(b) When a ray in the denser medium strikes the boundary at more than the critical angle, it is fully reflected back into the denser medium.

(c) Conditions: (1) light must go from a denser to a rarer medium; (2) the angle of incidence must exceed the critical angle.

(d) Examples: totally reflecting prisms in telescopes/periscopes, optical fibres, the sparkle of a diamond, a mirage.



Labelled diagram — schematic; parts as named in the answer. See the book diagram for exact form.

Question: Section (B) 8: Find the critical angle of diamond. The refractive index of diamond is 2.41.

Answer: sin C = 1 / 2.41 = 0.4149, so C = 24.5°.

Question: Section (B) 9: (a) What are optical fibres? (b) Describe total internal reflection in an endoscope.

Answer: (a) Optical fibres are thin strands of glass/plastic that carry light by total internal reflection.

(b) Light is sent down the fibre and keeps reflecting fully onward, so light is carried inside the body and an image is seen.

Question: Section (B) 10: (a) Describe image formation by a concave mirror. (b) How can a converging lens be used as a magnifying glass? (c) Describe a magnifying glass.

Answer: (a) Object beyond C: real, inverted, diminished image between C and F. At C: real, inverted, same-size image at C. Between C and F: real, inverted, enlarged image beyond C. At F: image at infinity. Inside F: virtual, erect, enlarged image behind the mirror.

(b) The object is placed inside the focal length, so a virtual, erect, enlarged image is seen.

(c) A magnifying glass is a convex lens of short focal length fitted in a frame/handle.

Question: Section (B) 11: Describe these optical instruments with ray diagrams and magnifying power: (i) simple microscope or magnifying glass (ii) compound microscope (iii) refracting telescope.

Answer: (i) Simple microscope: one convex lens;

object inside F, virtual, erect, enlarged image;

magnification M = 1 + D/f (D = 25 cm).

(ii) Compound microscope: the objective forms a real enlarged image, the eyepiece magnifies it further;

M = Mo x Me.

(iii) Telescope: an objective of long focal length forms a small real image, the eyepiece magnifies it;

M = fo/fe.

Question: Section (B) 12: Eye defects: (a) What do these terms mean? (i) Short sight (ii) Long sight. (b) How can these defects be corrected? (i) Short sight (ii) Long sight. (c) A normal eye cannot see objects closer than 25 cm clearly.

Answer: (a) (i) Short sight: distant objects are not seen clearly; the image forms in front of the retina. (ii) Long sight: near objects are not seen clearly; the image forms behind the retina.

(b) (i) Short sight: with a concave lens. (ii) Long sight: with a convex lens.

(c) The near point of a normal eye is 25 cm; the image of a closer object does not form sharply on the retina.

Question: Section (C) 1: A thin pin is placed 15 cm in front of a mirror of focal length 20 cm. Find the position and nature of the image.

Answer: By the book's method: 1/v = 1/f + 1/u = 1/20 + 1/15 = 7/60, so v = 8.57 cm.

[Note: with the usual sign convention, the image of an object inside the focal length of a concave mirror is virtual, 60 cm behind the mirror;

the book's printed answer follows its own 1/v = 1/f + 1/u working.]

Question: Section (C) 2: A mirror of focal length 13.5 cm forms an image when the object is 11.5 cm away. Find the distance of the image from the mirror in the same manner.

Answer: 1/v = 1/f + 1/u = 1/13.5 + 1/11.5 = 25/155.25, so v = 6.21 cm.

Question: Section (C) 3: A car's rear-view convex mirror has a radius of curvature of 4.0 m. If a bus stands 5.0 m from this mirror, find the nature and size of its image.

Answer: f = R/2 = 2.0 m.

1/v = 1/2 + 1/5 = 7/10, so v = 1.428 m behind the mirror.

Magnification m = 1.428/5 = 0.29 — a virtual, erect, diminished image.

[Note: the book prints 1.428 cm; the correct unit is m.]

Question: Section (C) 4: An object is placed 15 cm from a convex lens of focal length 10 cm. Find the position and size of its image.

Answer: By the book's method: 1/v = 1/f + 1/u = 1/10 + 1/15 = 1/6, so v = 6 cm;

magnification m = 6/15 = 0.4.

[Note: the book prints (2 cm);

2 matches the magnification under the standard lens formula (v = 30 cm,

m = 2).]

Question: Section (C) 5: A concave lens of focal length 20 cm forms an image 15 cm from the lens. Find the power of the lens. Also tell how far the lens is from the object. [The printed wording of the second part is unclear.]

Answer: Power P = 1/f = 1/0.20 m = 5 D (0.05 cm^-1).

Object distance (by the book's method): 1/u = 1/v - 1/f = 1/15 - 1/20 = 1/60, so u = 60 cm.

[Note: the book prints (0.05 cm) — 0.05 is 1/f in cm^-1, that is, 5 dioptres.]

Question: Section (C) 6: A ray enters water from air. The angle of incidence is 40°. If the refractive index of water is 1.33, find its angle of refraction.

Answer: sin r = sin 40° / 1.33 = 0.6428 / 1.33 = 0.4833, so r = 28.8°.

برق سکونی14یونٹ :

1. The branch of physics in which a static charge is studied is called _____.

(a) Electricity

✓ (b) Electrostatics

(c) Quantum

(d) Magnetism

2. When two identical point charges 1 metre apart repel each other with a force of 9x10^9 N, each charge is _____.

(a) Large

(b) 1 newton

(c) 2 newton

✓ (d) Equal to one coulomb

3. Coulomb's law closely resembles which law?

(a) The law of conservation of energy

✓ (b) Newton's law of gravitation

(c) Newton's second law of motion

(d) Faraday's law

Question: 4. If the electrostatic force between two electrons is (F) newton, what is the force between two protons at the same distance? (a) Large newton (b) Also (F) (c) 2/3 F newton (d) (F)

Answer: (b) Also (F) — the electron and the proton carry equal amounts of charge, so the force is equal. [Note: printed options (b) and (d) look identical.]

5. The direction of the electric force and the electric field intensity is _____.

(b) Perpendicular

(c) In the opposite direction

(d) Can be in any direction

6. The work done in moving a unit charge in an electric field is called _____.

(a) Electric field

(b) Electric current

✓ (c) Electric potential

(d) Electric field lines

7. In what way should capacitors be joined so that their capacitance increases?

✓ (a) Parallel

(b) Series

(c) Both

✓ (a) and

(b)

(d) None

8. If two 8 microfarad (8 µF) capacitors are joined in series, their equivalent capacitance will be _____.

(a) 1/4 microfarad

✓ (b) 4 microfarad

(c) 8 microfarad

(d) 16 microfarad

9. When a dielectric is present between the plates of a capacitor, its capacitance _____.

✓ (a) Increases

(b) Decreases

(c) Stays the same

10. If the area of the plates of a capacitor is increased, its capacitance

_____.

(a) Stays the same

(b) Becomes half

(c) Becomes double

(d) Becomes four times

Question: Section (B) 1: Explain how an electric charge can be detected with very simple experiments.

Answer: A comb or a plastic rod rubbed on hair, when brought near small bits of paper, attracts the bits — this shows that the rod carries a charge. In the same way, a rubbed balloon sticks to a wall.

Question: Section (B) 2: Explain how an electroscope works.

Answer: A charged object is touched to the cap of the electroscope, and the charge reaches the leaves. The same charge on both leaves makes them repel each other and open apart. The opening of the leaves shows the presence and amount of the charge.

Question: Section (B) 3: State and explain Coulomb's law.

Answer: The force between two point charges is directly proportional to the product of their amounts and inversely proportional to the square of the distance between them: F = k q1 q2 / r^2.

Like charges repel each other; opposite charges attract each other.

Question: Section (B) 4: State the electric field and the intensity of the electric field.

Answer: The electric field is the region around a charge in which a force is felt on another charge. The intensity of the electric field at a point is the force on a unit charge there: E = F/q, unit N/C.

Question: Section (B) 5: State the concept of electrostatic potential.

Answer: Electrostatic potential is the work done in bringing a unit positive

charge from infinity (the reference point) to a point: V = W/q, unit volt.

Question: Section (B) 6: State potential difference and how energy is transferred in moving a charge.

Answer: Potential difference is the work done in moving a unit charge between two points: V = W/q. Energy is transferred in moving a charge: W = qV — that much energy is gained by, or taken from, the charge.

Question: Section (B) 7: Give an example of the damage caused by electrostatic discharge when lightning strikes a building, and how to avoid it.

Answer: Lightning striking a building can set it on fire and harm people. Protection: a lightning conductor — a metal rod on top of the building joined to the ground by a thick wire; it carries the charge safely into the ground.

Question: Section (B) 8: State how a capacitor stores electric charge.

Answer: A capacitor has two conducting plates separated by an insulator. Charging one plate brings the opposite charge onto the other plate. The stored charge is Q = CV.

Question: Section (B) 9: Explain why it is necessary to find the equivalent capacitance of capacitors joined in series or in parallel.

Answer: So that a single capacitor giving the same effect can be used in place of the combination.

Series: 1/C = 1/C1 + 1/C2 + ...;

parallel: C = C1 + C2 + ...

Question: Section (B) 10: Give examples of how capacitors are used in different electrical appliances.

Answer: In fans (for starting), in radios/TVs (tuning circuits), in a camera

flash, in power supplies (for smoothing), in computers.

Question: Section (B) 11: In which direction does a charged particle move in an electric field?

Answer: A positive charge in the direction of the field, a negative charge opposite to the field — each accelerates in the direction of the force on it.

Question: Section (B) 12: Is the charge on each capacitor in a series combination always equal to the total charge collected by one capacitor?

Answer: Yes — in a series combination each capacitor carries the same charge, equal to the charge supplied by the battery.

Question: Section (C) 1: What is the electric force of repulsion between two electrons one metre apart?

Answer: F = k e^2 / r^2 = 9x10^9 x (1.6x10^-19)^2 / 1^2 = 2.3x10^-28 N.

Question: Section (C) 2: Two point charges q1 = 5 µC and q2 = 3 µC are placed 5 cm apart. Calculate the Coulomb force between them.

Answer: F = 9x10^9 x 5x10^-6 x 3x10^-6 / (0.05)^2 = 135x10^-3 / 0.0025 = 54 N.

Question: Section (C) 3: If the electric field on a 2 microcoulomb (2 µC) charge is 3.42x10^11 N/C, what is the force on it?

Answer: Force = qE = 2x10^-6 x 3.42x10^11 = 6.84x10^5 N.

Question: Section (C) 4: How much charge is on a capacitor? If a 40 microfarad (40 µF) capacitor has a potential difference of 6 volts?

Answer: Charge Q = CV = 40x10^-6 x 6 = 2.4x10^-4 C.

Question: Section (C) 5: The potential difference between two points is 100 V. If an unknown charge is moved between these points and the amount of work is 500 J, find the amount of charge.

Answer: Charge q = W/V = 500/100 = 5 C.

Question: Section (C) 6: Find the equivalent capacitance if 2 µF, 3 µF and 4

μF capacitors are joined in series.

Answer: 1/C = 1/2 + 1/3 + 1/4 = 13/12 (μF^-1), so C = 12/13 μF = 0.923 μF = 9.2x10^-7 F.

وهندڙ بجلي15يونٽ :

Question: Self Assessment Q1: Briefly explain the dangers of electricity at home.

Answer: Electric shock, burns and fire — touching a live wire through damaged insulation, overloading or bare wires lets current pass through the body.

Question: Self Assessment Q2: State four safety measures for a home circuit.

Answer: (1) Fit a fuse or circuit breaker.

(2) Fit an earth wire.

(3) Use good insulation and proper wiring.

(4) Do not overload sockets, and do not touch switches with wet hands.

Question: Self Assessment Q3: Does a fuse control voltage or current in a circuit?

Answer: Current — if the current rises above its rating, the fuse wire melts and breaks the circuit; it does not control voltage.

1. When electrons move from a lower potential to a higher potential in a circuit, they _____.

✓ (a) Gain energy

(b) Lose their existence

(c) Give out energy

(d) Gain potential

2. An ammeter is connected in an electric circuit by the _____ method.

(a) Series

(b) Parallel

(c) Any

(d) None of these

3. Resistance does not depend on which of the following?

(a) Length of the conductor

(b) Cross-sectional area

✓ (c) Density

(d) Resistivity

4. Ohm's law states that _____.

(a) Current increases as resistance increases

(b) Current increases as resistance decreases

(c) Voltage increases with resistance

✓ (d) Voltage increases as current increases

5. The state of a circuit in which the resistance is very large is called _____.

(a) Complete circuit

(c) Short circuit

(d) Second circuit

6. For Ohm's law to apply, the condition is _____.

✓ (a) The temperature must stay constant

(b) Current must be directly proportional to voltage

(c) There is resistance of a round wire

(d) Matyan sup

7. Ohm's law does not apply to _____.

✓ (a) Semiconductors

(b) DC circuits

(c) Small resistances

✓

(d) Large current

8. When 6Ω and 12Ω resistances are joined in parallel, their equivalent resistance is:

(a) 7Ω

(b) 6Ω

✓ (c) 4Ω

(d) 5Ω

9. The property of a body that opposes the flow of electric charges is called electric _____.

(b) Potential

✓ (c) Resistance

(d) Conduction

10. What is the reason for connecting an electric circuit to a battery?

(a) To keep resistance constant in the conductor

(b) To change resistance in the conductor

✓ (c) To maintain a potential difference in the conductor

(d) To maintain a changing potential in the conductor

Question: Section (B) 1: Does a series combination of capacitors store an equal amount of charge in each capacitor?

Answer: Yes — in a series combination each capacitor carries the same charge, equal to the charge supplied by the battery.

Question: Section (B) 2: Why do we prefer joining more appliances in parallel rather than in series? What are the advantages of this arrangement?

Answer: Because in parallel each appliance gets the full voltage, the others keep working if one fails, and each appliance can be switched on/off separately.

Question: Section (B) 3: Is a potential difference needed to pass current

through a circuit?

Answer: Yes — the potential difference is what pushes the charges. Without a potential difference no current flows in the circuit.

Question: Section (B) 4: Can a bulb and an electric heater be joined in series? What happens?

Answer: They can be joined, but the voltage is shared between them — the bulb glows dimly and the heater does not heat fully. That is why household appliances are joined in parallel.

Question: Section (B) 5: When a fuse is used in a circuit, does it control the current or the potential difference?

Answer: The current — if the current rises above its rating, the fuse wire melts and breaks the circuit. It does not control voltage.

Question: Section (B) 6: What is meant by conventional current? Explain.

Answer: Conventional current is the current taken to flow from the positive terminal toward the negative terminal — the direction of flow of positive charge. It is opposite to the flow of electrons.

Question: Section (B) 7: State Ohm's law and its limits.

Answer: Ohm's law: at constant temperature, the current through a conductor is directly proportional to the potential difference across its ends: V = IR. Limits: it applies only to ohmic conductors; it does not apply when the temperature changes, or to non-ohmic devices such as semiconductors/diodes.

Question: Section (B) 8: Find the equivalent resistance of different resistances using a suitable solution in series or parallel combination.

Answer: Series: R = R1 + R2 + R3 + ...

(the equivalent is larger than even the largest resistance).

Parallel: 1/R = 1/R1 + 1/R2 + 1/R3 + ...

(the equivalent is smaller than even the smallest resistance).

Question: Section (B) 9: What factors affect the resistance of a metal conductor, and how are they measured?

Answer: Factors: length (longer = more resistance), area (thicker = less resistance), material (resistivity), temperature (in metals, resistance rises with temperature). Measurement: directly with an ohmmeter, or from V/I using a voltmeter and an ammeter.

Question: Section (B) 10: Explain Joule's law and the energy wasted in resistances.

Answer: Joule's law: the heat produced in a resistance is H = I^2Rt — directly proportional to the square of the current, the resistance and the time.

In a resistance, electrical energy is converted into heat and wasted.

Question: Section (B) 11: Explain the live, neutral and earth wires in a household circuit.

Answer: Live wire: carries the current, at 220 V. Neutral wire: the return path, at about 0 V. Earth wire: a safety wire joined to the ground; it carries current only in a fault.

Question: Section (B) 12: How does alternating current work?

Answer: In alternating current (AC) the direction of the current keeps reversing — in Pakistan 50 times per second — and the voltage changes in the same way. It is produced by generators and supplied to homes.

Question: Section (B) 13: Explain the dangers associated with electric current (danger of insulators, overheating of wires).

Answer: Electric shock and burns (touching a bare wire or damaged

insulation), fire from overheated wires (on overload or short circuit), and damage to appliances.

Question: Section (B) 14: How are safety measures used in household electricity? Explain.

Answer: Fuse/circuit breaker (breaks the circuit on excess current), earth wire (sends fault current into the ground), good insulation, proper wiring, not overloading sockets.

Question: Section (B) 15: State the cause of the effect of an appliance's electricity on the human body.

Answer: Current passing through the body affects the nerves and muscles — even a small current is felt; a larger current causes burns, muscle contraction and disturbance of the heartbeat.

Question: Section (C) 1: When a pocket calculator carries a current of 0.0002 A, how much charge flows in each minute?

Answer: Charge Q = It = 0.0002 x 60 = 0.012 C = 12 mC.

Question: Section (C) 2: Find the amount of current used by an electric heater to heat a room in 5 minutes, when the charge is 2100 C.

Answer: Current I = Q/t = 2100 / 300 = 7 A.

Question: Section (C) 3: The potential difference between two points is 90 V. If an unknown charge is moved between these points and the amount of work is 450 J, find the amount of charge.

Answer: Charge q = W/V = 450/90 = 5 C.

Question: Section (C) 4: Find the potential difference between two points (A) and (B). The external work done in moving a +9 µC charge from (A) to (B) is 9x10^-4 J.

Answer: V = W/q = 9x10^-4 / 9x10^-6 = 100 V.

[Note: the printed work figure is unclear in the scan; the 100 V answer

implies 9x10^-4 J.]

Question: Section (C) 5: A portable radio's battery is at a potential of (6.0 V) when a current of 20 milliamperes is given to the radio circuit. Find the resistance of the radio.

Answer: R = V/I = 6.0 / 0.020 = 300 Ω.

Question: Section (C) 6: The possible difference applied across the terminals of a six-volt battery is 6.0 volts. Estimate the resistance of the radio when the current flowing through it is 20 mA.

Answer: R = V/I = 6.0 / 0.020 = 300 Ω.

(Same as question 5.)

Question: Section (C) 7: Resistances 4Ω, 6Ω, 12Ω are joined in parallel and connected to a 6 V (EMF) source. Find: (i) the equivalent resistance of the circuit (ii) how much current flows through the circuit (iii) how much current flows through each resistance.

Answer: (i) 1/R = 1/4 + 1/6 + 1/12 = 6/12 = 1/2, so R = 2 Ω.

(ii) Current I = V/R = 6/2 = 3 A.

(iii) I1 = 6/4 = 1.5 A;

I2 = 6/6 = 1 A;

I3 = 6/12 = 0.5 A.

Question: Section (C) 8: In a 220 V circuit, 80 Watt and 120 Watt bulbs can be compared. Tell which bulb has greater R and through which bulb more current passes.

Answer: R = V^2/P: the 80 W bulb's R = 220^2/80 = 605 Ω;

the 120 W bulb's R = 220^2/120 = 403 Ω.

So the 80 W bulb has the greater R, and more current passes through the 120 W bulb (120/220 = 0.545 A, against 80/220 = 0.364 A).

برقي مقناطيسيت16يونٹ :

Question: Self Assessment Q1: State mutual induction.

Answer: When a change of current in one coil produces an emf in a nearby second coil, it is called mutual induction.

Question: Self Assessment Q2: List the factors affecting the induced e.m.f.

Answer: (1) Strength of the magnet.

(2) Speed of motion.

(3) Number of turns in the coil.

(4) Area of the coil.

Question: Self Assessment Q3: How does an AC generator work?

Answer: A coil rotates in a magnetic field, so the flux through it changes and a changing emf is produced; the current reverses direction after every half turn — that is AC. Slip rings carry it out.

Question: Self Assessment Q4: What is a transformer and how does it work?

Answer: A transformer is a stationary machine that increases or decreases AC voltage, on the principle of mutual induction. AC in the primary produces a changing flux in the iron core, which induces an emf in the secondary.

Question: Self Assessment Q5: What is the difference between a step-up and a step-down transformer?

Answer: Step-up: more turns in the secondary, voltage increases.

Step-down: fewer turns in the secondary, voltage decreases.

Question: Self Assessment Q6: For what work are step-up and step-down transformers used in power transmission?

Answer: At the power plant a step-up transformer raises the voltage very high so that heat loss in long-distance transmission is small; in the neighbourhood a step-down transformer lowers it to 220 V for homes.

Question: 1. Which statement about magnetic poles is correct? (a) Like poles repel each other (b) Unlike poles attract each other (c) Magnetic poles do not affect each other (d) A single magnetic pole cannot exist

Answer: (d) A single magnetic pole cannot exist — poles always occur in pairs. [Note:

(a) and (b) are also true.]

2. What is the direction of the magnetic lines inside a bar magnet?

(a) From the north pole to the south pole

✓ (b) From the south pole to the north pole

(c) From side to side

(d) There are no magnetic lines there

3. Through what can the presence of a magnetic field be felt?

(a) A small needle

(b) A static charge

(c) A static charge cutter

✓ (d) A compass

4. If the current in a wire placed perpendicular to a magnetic field is increased, the force on the wire _____.

✓ (a) Increases

(b) Decreases

(c) Stays the same

(d) Becomes zero

5. A D.C. motor converts _____.

(a) Mechanical energy into electrical energy

(b) Mechanical energy into chemical energy

✓ (c) Electrical energy into mechanical energy

(d) Electrical energy into chemical energy

6. Which part of a D.C. motor reverses the current in the coil after every half cycle?

(a) Armature

✓ (b) Commutator

(c) Brush

(d) Rings

7. Induced emf in a circuit is consistent with the law of conservation of _____.

(a) Mayu

(b) Charge

(c) Momentum

✓ (d) Energy

8. A step-up transformer _____. (a) Increases the input current (b) Increases the input voltage (c) Has more turns in the primary (d) Has more turns in the secondary coil

9. A transformer has a turns ratio of 10, meaning _____.

(a) Is = 10 Ip

(b) Np = 10 Ns

✓ (c) Ns = 10 Np

(d) Vs = 10 Vp

Question: Section (B) 1: A magnetic field produces voltage in a wire. In which direction should the wire be moved relative to the magnetic field to produce maximum voltage?

Answer: Perpendicular to the field (90°) — the wire must cut the magnetic lines at a right angle. Motion along the field produces no voltage.

Question: Section (B) 2: Can a transformer work on D.C. current?

Answer: No — with D.C. the field stays steady, the flux does not change, so no emf is induced in the secondary (only a momentary pulse at switch-on/off).

Question: Section (B) 3: Explain with an experiment how current is produced in a conductor around a magnetic field.

Answer: Join a coil to a galvanometer. Moving a magnet toward/away from the coil deflects the needle — an induced current is produced. Stopping the magnet brings the needle to zero. The faster the motion, the greater the current.

Question: Section (B) 4: Explain how a force acts on a current-carrying conductor when the conductor is perpendicular to the magnetic field.

Answer: The force on the wire is maximum: F = BIL. Its direction is given by Fleming's left-hand rule: thumb = force, forefinger = field, middle finger = current.

Question: Section (B) 5: State that a current-carrying coil in a magnetic field feels a torque.

Answer: Opposite forces on the two sides of the coil give a turning torque: $\tau = NIAB \sin\theta$. This is the basis of the motor.

Question: Section (B) 6: Describe an experiment showing emf produced in a circuit by a changing magnetic field.

Answer: Join a coil to a galvanometer and move a magnet in/out — the needle deflects, meaning an induced emf is produced. Stopping the magnet ends the emf.

Question: Section (B) 7: Give examples of things that can increase or decrease the strength of the induced emf.

Answer: Increase it: a stronger magnet, faster motion, more turns, larger area. Decrease it: a weaker magnet, slower motion, fewer turns, smaller area.

Question: Section (B) 8: Explain that an induced emf opposes the change that produces it. Show the relation with the rule.

Answer: Lenz's law: the induced current flows in a direction that opposes the change that produced it — e.g. bringing a magnet toward a coil makes the near end of the coil the same pole, repelling it. This is consistent with the conservation of energy.

Question: Section (B) 9: Explain how an A.C. generator works in its simplest form.

Answer: A coil rotates in a magnetic field; the changing flux through it produces an emf. The current reverses direction after every half turn, so the output is alternating current. Slip rings and brushes carry it to the outer circuit.

Question: Section (B) 10: State mutual induction and give an example.

Answer: A change of current in one coil produces an emf in a nearby second coil. Example: a transformer — the changing current in the primary induces an emf in the secondary.

Question: Section (B) 11: Show that a transformer works on the concept of mutual induction.

Answer: AC in the primary produces a changing magnetic flux in the iron core. That flux passes through the secondary and induces an emf in it — that is mutual induction.

Question: Section (B) 12: What functions do transformers perform in A.C. circuits?

Answer: They raise/lower the voltage — raising it for transmission, lowering it for use; they also change the current in inverse proportion.

Question: Section (B) 13: Describe the role of transformers in the flow of electric current from the power plant to your home.

Answer: At the power plant a step-up transformer raises the voltage very high (so loss is small). Electricity travels through long wires. In the neighbourhood a step-down transformer lowers the voltage to 220 V for homes.

Question: Section (B) 14: Make a list of uses of step-up and step-down transformers found in your home.

Answer: Step-down: mobile chargers, adapters, doorbells, toy transformers. Step-up: stabilizers, UPS/inverters, microwave ovens.

Question: Section (C) 1: A wire carrying a current of 4 A is 15 cm long. It is placed at 30° in a uniform magnetic field of 0.8 T between the two poles of a magnet. Find the force acting on the wire.

Answer: Force F = BIL sinθ = 0.8 x 4 x 0.15 x sin30° = 0.8 x 4 x 0.15 x 0.5 = 0.24 N.

Question: Section (C) 2: A square wire coil of side 2.0 cm carries a current of 2.0 A. The plane of the coil makes an angle of 30° with a magnetic field of 0.7 T. What is the magnitude of the torque on the coil? (Torque)

Answer: Area A = 0.02 x 0.02 = 4x10^-4 m^2.

The plane makes 30° with the field, so the angle with the normal is 60°.

Torque τ = NIAB sin60° = 1 x 2 x 4x10^-4 x 0.7 x 0.866 = 4.8x10^-4 Nm.

[Note: the scan shows the side as "20.cm", i.e. 2.0 cm; 20 cm does not give this answer.]

Question: Section (C) 3: A transformer is required to convert 220 V to 12 V. If the primary coil has 2200 turns, find the number of turns in the

secondary coil.

Answer: Ns/Np = Vs/Vp, so Ns = 2200 x 12/220 = 120 turns.

Question: Section (C) 4: A coil is wound around a solenoid in which the current is changing at the rate of 150 A/s. The mutual inductance of the two coils is 5.5x10^-5 H. Find the induced emf in the secondary coil.

Answer: emf = -M ΔI/Δt = -5.5x10^-5 x 150 = -8.25x10^-3 V.

(The minus sign shows the emf opposes the change.)

تعارفي اليکٹرانکس17یونٹ :

Question: Self Assessment Q1: What is a logic gate?

Answer: A logic gate is a digital circuit that makes a logical decision on one or more inputs and gives one output — e.g. AND, OR, NOT gates.

Question: Self Assessment Q2: Which gate uses an inverted input?

Answer: The NOT gate (inverter) — it inverts the input: it makes 0 into 1 and 1 into 0. NAND and NOR gates also include inversion.

Question: Self Assessment Q3: Write the Boolean expression of an OR gate.

Answer: Y = A + B — the output is 1 when any one input is 1.

1. The charge carriers in a metallic conductor are free:

✓ (a) Electrons

(b) Protons

(c) Neutrons

(d) Photons

2. A continuous flow of electrons is made possible by the ___ of a device.

(a) Cathode

(b) Electron gun

(c) Anode

✓ (d) Filament

3. An electric field can be detected by a _____.

(a) Proton

(b) Neutron

(c) Forces

✓ (d) Electron

4. If the direction of the magnetic field is reversed, the direction of the

force is:

✓ (a) Reversed

(b) The same

(c) Reversed and straight

(d) None of these

5. The process of emission of electrons from a hot metal is called:

(a) Emission of plastic

✓ (b) Thermionic emission

(c) Emission of current

(d) Emission of state

Question: 6. If the input of a NOT gate is "1", what is its output? (a) 1 (b) Zero (0) (c) 1 or zero (d) None of these

Answer: (b) Zero (0) — the NOT gate inverts the input: the output of 1 is 0.

[Note: printed option (b) reads "wadi", taken as zero (0).]

7. The Boolean expression of an AND gate is:

✓ (a) A.B

(b) A + B

(c) A x B

(d) None of these

8. Electronics consists of:

(a) Physics

(b) Engineering

(c) Technology

✓ (d) All of these

9. The Boolean expression of an OR gate is:

(a) A.B

✓ (b) A + B

(c) A x B

(d) None of these

10. A cathode ray is:

(a) Positive charge

(b) Negative charge

(c) Positron

(d) No charge

Question: Section (B) 1: Show with an example that the world is shifting from low-technology appliances to higher-technology appliances.

Answer: Example: LED/LCD TVs in place of old CRT TVs, smartphones in place of simple mobiles, and computerised billing in place of cash registers — all because of digital electronics.

Question: Section (B) 2: (a) Write any advantages of digital electronics devices over analogue electronics devices. (b) State the role of the vacuum tube in electronics.

Answer: (a) Digital devices are more accurate, noise-free, smaller, faster, and can store information.

(b) The vacuum tube controlled the flow of electrons; it was used in early radios, TVs, amplifiers and computers.

Question: Section (B) 3: What is the function of a DAC?

Answer: A DAC (digital-to-analogue converter) changes a digital signal into an analogue signal — e.g. a computer's digital sound into an analogue signal for a speaker.

Question: Section (B) 4: What makes it possible for the cathode to emit electrons?

Answer: Heating — the filament heats the cathode so that the electrons get enough energy to escape from the metal surface (thermionic emission).

Question: Section (B) 5: (a) Explain the process of thermionic emission with a diagram. (b) If instead of a vacuum there were air in the vacuum tube, would thermionic emission still be possible?

Answer: (a) The hot filament heats the cathode; the electrons gain energy and come out of the metal surface and are attracted toward the anode.

(b) Emission would still occur, but the electrons would collide with air molecules and the cathode would burn — the tube would not work properly.

Question: Section (B) 6: Why are electrons attracted toward the anode?

Answer: Because the anode is positive and attracts the negative electrons — the potential difference between them accelerates the electrons.

Question: Section (B) 7: Write any two properties of cathode rays.

Answer: (1) They travel in straight lines.

(2) They carry negative charge and are deflected in electric and magnetic fields.

Question: Section (B) 8: (a) If the gas in the tube is changed, will the properties of cathode rays change? (b) If the material of the cathode is changed, will the cathode rays change? (c) Name any two instruments that use an electron beam. (d) Through cathode rays, which particle was discovered?

Answer: (a) No — cathode rays are the same (electrons) in every gas.

(b) No — the same rays are obtained with every cathode material.

(c) Cathode-ray oscilloscope (CRO) and TV (CRT) / electron microscope.

(d) The electron.

Question: Section (B) 9: Explain the working of an electron gun and how it produces the electron beam.

Answer: The filament heats the cathode → thermionic emission releases electrons → the positive anode accelerates them into a beam → control and focusing grids shape the beam.

Question: Section (B) 10: If an electron beam passes through an electric field, will there be a change in its direction? Explain.

Answer: Yes — the beam bends toward the positive plate, because the electrons are negative and the positive plate attracts them.

Question: Section (B) 11: How is the electron beam produced by an electron gun directed toward a specific target?

Answer: Deflecting plates (electric field) or deflection coils (magnetic field) bend the beam up/down and left/right toward the target.

Question: Section (B) 12: Explain with a diagram the bending of electron beams by an electric field.

Answer: The beam passes between two charged plates and bends toward the positive plate. The bending depends on the strength of the field, the charge of the electron and its speed.

Question: Section (B) 13: (a) When an electron beam passes through a magnetic field, what effect is there on its energy? (b) When an electron beam passes through a magnetic field, does its speed change?

Answer: (a) No effect — the magnetic force is perpendicular to the motion and does no work, so the energy stays the same.

(b) No — the speed stays the same; only the direction changes.

Question: Section (B) 14: Describe the effect produced on the bending of electron beams passing through a magnetic field.

Answer: The beam bends in a circular path; the direction follows Fleming's left-hand rule. The radius of the bend depends on the speed, the field, the charge and the mass.

Question: Section (B) 15: Explain the following parts of an oscilloscope: (a) Fluorescent screen (b) Cathode (c) Anode (d) Y plate.

Answer: (a) Fluorescent screen: it glows where the beam strikes and shows the picture.

(b) Cathode: heated, it emits electrons.

(c) Anode: it accelerates the electrons into a beam.

(d) Y plate: it deflects the beam up/down (vertical deflection).

Question: Section (B) 16: Explain how the electron beam is produced inside a cathode-ray oscilloscope.

Answer: The filament heats the cathode → thermionic emission releases electrons → the anode accelerates them into a beam → grids focus the beam.

Question: Section (B) 17: Explain the term "Logic" with a suitable example.

Answer: Logic means a decision based on conditions. Example: a car's alarm sounds when the ignition is on AND the seat belt is not fastened — both conditions together switch the alarm on.

Question: Section (B) 18: (a) Name the components used for logic in digital electronics. (b) What is the cause of acceleration of electrons flowing from cathode to anode? Explain.

Answer: (a) Logic gates (AND, OR, NOT, NAND, NOR).

(b) The potential difference between cathode and anode — the positive

anode attracts and accelerates the electrons.

Question: Section (B) 19: Explain why either of the two states (0 or 1) of a digital signal can exist.

Answer: A digital circuit has two stable states — on/high (1) and off/low (0). The output is always one of the two.

Question: Section (B) 20: (a) Draw the symbol of a NAND gate. (b) Draw the truth table of an AND gate.

Answer: (a) NAND symbol: the AND symbol with a small circle (bubble) at the output.

(b) AND truth table: A=0,B=0 → Y=0;

A=0,B=1 → Y=0;

A=1,B=0 → Y=0;

A=1,B=1 → Y=1.

Question: Section (B) 21: (a) Explain the logic operation of an inverter. (b) Draw the truth table of an OR gate.

Answer: (a) An inverter (NOT gate) inverts the input: 0 → 1, 1 → 0.

(b) OR truth table: A=0,B=0 → Y=0;

A=0,B=1 → Y=1;

A=1,B=0 → Y=1;

A=1,B=1 → Y=1.

Question: Section (B) 22: Which two logic gates give output (0) with inputs (1) and zero (0)? (a) Draw the symbol of a NOR gate. (b) Draw the truth table of a NOR gate.

Answer: The AND gate and the NOR gate — both give output 0 for (1,0).

(a) NOR symbol: the OR symbol with a small circle (bubble) at the output.

(b) NOR truth table: A=0,B=0 → Y=1;

A=0,B=1 → Y=0;

A=1,B=0 → Y=0;

A=1,B=1 → Y=0.

[Note: the first part of the question is unclear in the scan; taken as the two gates giving 0 for (1,0).]

معلومات مواصلات ٹیکنالاجي18یونٹ :

1. The other name of a supercomputer is:

(a) A computer that works less

(b) Maker-mem computer

(c) Mainframe computer

✓ (d) None

2. Input, processing, output and storage are collectively referred to as:

✓ (a) The information cycle

(b) The software life cycle

(c) The hardware life cycle

(d) Information technology

3. The input and output of a computer are called:

✓ (a) Data

(b) Information

(c) Computer

(d) Mouse

4. Which of the following is not a system software?

(a) Assembler

(b) Interpreter

(c) Compiler

✓ (d) Tally

5. Which is the best way to connect different computers in office buildings?

(a) MAN

(b) WAN

(c) ANN

(d) LAN

6. A computer program that translates computer programs into machine language at the same time is called:

✓ (a) Interpreter

(b) CPU

(c) Compiler

(d) Simulator

7. A sequence of instructions in computer language given a name to achieve a desired result is:

✓ (a) A program

(b) A decision table

(c) Pseudo Code

(d) An algorithm

8. USB stands for:

(a) Ultra Serial Bus

(b) Unlimited Structured Bit

✓ (c) Universal Serial Bus

(d) Unified States Bus

9. Which extension is not appropriate in MS-Word?

(a) .doc

(b) .docx

(c) .rtf

✓ (d) .jpeg

10. What does ICT stand for?

✓ (a) Information and Communication Technology

(b) Integrated Circular Technology

(c) Intensive Computer Techniques

(d) Interfacing Computer Theories

✓

Question: Section (B) 1: What is the difference between data and information?

Answer: Data is raw figures/facts with no meaning by themselves; information is that data after processing has made it meaningful — e.g. students' marks are data, the result is information.

Question: Section (B) 2: What do you understand by Information and Communication Technology (ICT)?

Answer: ICT is the technology used to store, process and send information from one place to another — computers, the internet, mobile phones, radio, TV, etc.

Question: Section (B) 3: What are the parts of information technology? Show the function of each part.

Answer: (1) Input — putting data in (keyboard, mouse).

(2) Processing — calculating/deciding on data (CPU).

(3) Output — showing the result (monitor, printer).

(4) Storage — keeping data safe (hard disk).

(5) Communication — sending information (internet).

Question: Section (B) 4: State the difference between primary memory and secondary memory.

Answer: Primary memory (RAM, ROM) is inside the computer, fast and small; RAM is emptied when the power is switched off. Secondary memory (hard disk, CD) is external, slower, larger and permanent.

Question: Section (B) 5: Name different information storage devices and state their uses.

Answer: Hard disk (large permanent storage), CD/DVD (films/software), USB flash (carrying data from place to place), memory card (mobile/camera), floppy disk (small old storage).

Question: Section (B) 6: Explain the transmission of radio waves through space.

Answer: Radio waves travel through space at the speed of light. A transmitter sends the signal, an antenna spreads it into space, and a receiver picks it up and turns it into sound/picture.

Question: Section (B) 7: How are light signals sent through optical fibres?

Answer: Light travels forward inside the fibre by repeated total internal reflection. The signal arrives fast, with little loss and in large quantity.

Question: Section (B) 8: What is a computer? What is the importance of computers in daily life?

Answer: A computer is a machine that takes input, processes it, gives output and stores data. In daily life it is used in education, business, banking, health, communication and entertainment.

Question: Section (B) 9: What is the difference between hardware and software? Name different software.

Answer: Hardware is the physical parts of the computer (keyboard, monitor, CPU); software is the instructions/programs that run the hardware. Software: system software (Windows, Linux), application software (MS Word, browsers).

Question: Section (B) 10: What do you understand by word processing and data management?

Answer: Word processing: writing, correcting and printing documents (MS Word). Data management: storing, organising and retrieving data when needed (databases).

Question: Section (B) 11: What is the internet? Discuss internet,

information and usefulness.

Answer: The internet is the worldwide network of computers. It is the greatest source of information — useful for education, research, business, communication and entertainment.

Question: Section (B) 12: Discuss the role of information technology in school education.

Answer: E-learning, online classes, digital books, internet for research, smart classrooms, management of results and records — it makes study easier and more interesting.

Question: Section (B) 13: Why are optical fibres more useful tools in communication?

Answer: Because they are fast, carry much information, lose little signal, are safe from electrical interference, and are light and thin.

Question: Section (B) 14: Which is more reliable, a floppy disk or a hard disk?

Answer: The hard disk — its capacity is far greater, its speed faster, its life longer, and data is safer on it.

Question: Section (B) 15: What is the difference between (RAM) and (ROM) memories?

Answer: RAM is temporary — it can be written and read, and is emptied when the power is switched off. ROM is permanent — it can only be read, and keeps its data even when the power is off.

ائٽم جي بناوت19يونٽ :

Question: Self Assessment Q1: In an atom there are electrons revolving around the nucleus, and protons and neutrons in the centre. Tell which of them (i) carry equal and opposite charges (ii) have nearly equal masses.

Answer: (i) The electron and the proton carry equal and opposite charges (electron negative, proton positive).

(ii) The proton and the neutron have nearly equal masses.

Question: Self Assessment Q2: Uranium-235 nuclide is denoted 92U^235. Find the proton number, electron number, neutron number and nucleon number of uranium.

Answer: Proton number Z = 92;

electron number = 92 (in a neutral atom);

neutron number N = A - Z = 235 - 92 = 143;

nucleon number A = 235.

1. 1H^2 and 1H^3 are:

✓ (a) Isotopes

(b) Isobars

(c) Isotones

(d) Isochores

2. In the neutral atoms of all isotopes of an element there is:

(a) A different number of protons

(b) The same number of neutrons

(c) The same number of protons

(d) The same number of nucleons

3. The two isotopes of chlorine are 17Cl^35 and 17Cl^37; in them the number of _____ is the same.

(a) Nucleons

✓ (b) Protons

(c) Neutrons

(d) Mass number

4. The atomic mass of an element is equal to _____.

✓ (a) The combined mass of protons and neutrons

(b) The combined mass of protons and electrons

(c) The combined mass of electrons and neutrons

(d) The mass of protons only

5. Most of the mass of an atom is found in the _____.

✓ (a) Nucleus

(b) Neutrons

(c) Protons

(d) Electrons

6. Suppose the isotope of uranium is 92U^237. The number of protons in it is _____.

✓ (a) 92

(b) 237

(c) 145

(d) 329

7. The symbol for proton number is:

(a) P

(b) A

(c) N

✓

(d) Z

Question: 8. The number of neutrons in protium is: (a) Zero (0) (b) One (c) Two (d) Three

Answer: (a) Zero (0) — the nucleus of protium (^1H) has only one proton, no neutron. [Note: printed option (a) reads "wadi", taken as zero (0).]

9. Compared with the rest of the atom, the nucleus is _____.

(a) Large in size and large in mass

✓ (b) Small in size and large in mass

(c) Large in size and small in mass

(d) Small in size and small in mass

10. If an element B consists of five protons and six neutrons, what is its symbol?

(a) 6B^11

(b) 11B

✓ (c) 5B^11

(d) 11B^5

Question: Section (B) 1: (a) Which particles are present in the nucleus of an atom? (b) State the structure of the atom. (c) How does the number of protons in the nucleus explain the difference between two elements?

Answer: (a) Protons and neutrons (nucleons).

(b) At the centre of the atom is a small, heavy, positive nucleus; electrons revolve around it; most of the atom is empty.

(c) The number of protons (atomic number) is distinct for each element — two elements differ in their proton numbers, so they are different elements.

Question: Section (B) 2: (a) Explain the Geiger-Marsden experiment with a

✓

diagram. (b) Write Rutherford's atomic model. (c) Why was it suggested that most of the atom is empty?

Answer: (a) Alpha particles (positive) were fired at a thin gold foil: most passed straight through, some deflected, very few bounced back.

(b) Rutherford's model: the mass and positive charge of the atom are in a small central nucleus; electrons revolve around it; most of it is empty.

(c) Because most alpha particles passed straight through without deflecting — meaning most of the atom is empty.

Question: Section (B) 3: (a) State the atomic number Z. (b) How is an atom of an element shown by a symbol?

Answer: (a) The atomic number Z is the number of protons in the nucleus (in a neutral atom the number of electrons is the same).

(b) Symbol: zX^{A} — X is the symbol of the element, Z at lower left (atomic number), A at upper left (mass number). Example: $_{6}C^{12}$.

Question: Section (B) 4: (a) What are isotopes? (b) Explain with examples of isotopes. (c) Why are the chemical properties of different isotopes of an element the same? (d) Name a physical property of an element that differs from one another.

Answer: (a) Isotopes are atoms of the same element with equal protons but different neutrons.

(b) Examples: hydrogen's ^{1}H, ^{2}H (deuterium), ^{3}H (tritium); carbon's ^{12}C and ^{14}C; chlorine's ^{35}Cl and ^{37}Cl.

(c) Chemical properties depend on electrons, and isotopes have equal

electrons.

(d) Mass (mass number) and related properties such as density — isotopes have different masses.

نیوکلیئر بناوت20یونٹ :

1. Alpha (α) radioactive rays are:

(a) A stream of fast-moving electrons

(b) A type of electromagnetic radiation

(c) More penetrating than beta rays

✓ (d) More ionizing than gamma rays

2. When a radioactive nuclide emits beta (β) particles, the atomic number (proton number) of the nucleus _____.

(a) Does not change

✓ (b) Increases by one

(c) Decreases by two

(d) Decreases by four

3. A radioactive element emits a particle from its nucleus consisting of two protons and two neutrons. The name of this process is:

✓ (a) Alpha emission

(b) Beta emission

(c) Gamma emission

(d) Nuclear fission

4. A radioactive decay equation is given: 91Pa^233 → 92U^233 + ...

(a) Gamma ray

(b) Proton

(c) Alpha particles

✓ (d) Beta particles

5. The common type of radioactive ray that moves in a straight line in an electric field is _____.

(a) Proton

(b) Electron

(c) Alpha particle

✓ (d) Gamma ray

6. A powder contains 100 mg of radioactive material emitting alpha particles. The half-life of the isotope is five days. After ten days, the remaining amount of the isotope will be:

(a) Zero

✓ (b) Twenty-five milligrams

(c) Fifty grams

(d) Fifty milligrams

7. The main source of energy in stars is:

(a) Chemical reaction

(b) Nuclear fission

✓ (c) Nuclear fusion

(d) Mechanical

8. The splitting of a heavy nucleus into two smaller nuclei is called:

(a) Fusion

✓ (b) Fission

(c) Half-life

(d) Gamma decay

9. The process in which two light nuclei join to form a heavy nucleus is called:

✓ (a) Nuclear fusion

(b) Nuclear fission

(c) Beta decay

(d) Alpha decay

10. Which row below correctly shows the nature of beta particles and the material that stops them?

(a) Helium nucleus | Through a few-millimetre sheet of aluminium

(b) Helium nucleus | Through a thin sheet of paper

✓ (c) Electron | Through a few-millimetre sheet of aluminium

(d) Electron | Through a thin sheet of paper

Question: 11. Which statement about gamma rays is correct? (a) They are the main type of radioactive ray that carries charge (b) Their ionizing effect is the clearest of all (c) Their penetrating power is the clearest of all (d) Their mass is very small (negligible)

Answer: (c) Their penetrating power is the clearest of all — gamma rays penetrate the most. [Note:

(d) is also true — gamma photons have no rest mass.]

12. Radiation exposure causes severe damage to health:

(a) Cancer

(b) Genetic mutation

(c) Deep burns

✓ (d) All of these

Question: 13. Radioactive materials must be handled with care. Which safety measures reduce the dangers during the use of radioactive materials? (a) Keep the materials at a distance (b) Keep the materials at a low temperature (c) Use a glass screen (d) Use the materials for a short time

Answer: (a) Keep the materials at a distance — increasing the distance reduces the effect of the rays. [Note:

(d) short time is also one of the book's safety measures.]

14. A scientist carried out an experiment using a sealed radioactive source emitting beta particles.

(a) Hold the material with tongs

(b) Keep the materials at a low temperature

(c) Open the laboratory windows

(d) Wash hands on leaving the laboratory

15. What is the safe way to dispose of a large amount of radioactive waste?

(a) Bury it inside a mountain

(b) Dissolve it and throw it in a drain

(c) Burn it in fire

(d) Flush it into the sea

Question: Section (B) 1: (a) Explain the term radioactivity. (b) What is meant by an unstable nucleus? (c) Why are some elements radioactive and some not?

Answer: (a) Radioactivity: the spontaneous emission of alpha, beta or gamma rays from an unstable nucleus.

(b) An unstable (spontaneous) nucleus is one that breaks up by itself and emits rays.

(c) Nuclei whose proton/neutron ratio is out of balance are unstable and radioactive; balanced nuclei are stable.

Question: Section (B) 2: The uranium isotope 92U^238 has atomic mass 238 and atomic number 92. (a) What is the nucleon number? (b) The uranium-238 nuclide emits an alpha particle and changes into a thorium (Th) nuclide. Tell: (i) the nucleon number of the alpha particle (ii) the proton number of thorium (iii) the thorium nuclide (c) Complete the nuclear equation of uranium decay.

Answer: (a) Nucleon number A = 238.

(b) (i) Nucleon number of the alpha particle = 4.

(ii) Proton number of thorium = 92 - 2 = 90.

(iii) Thorium nuclide = 90Th^234.

(c) 92U^238 → 90Th^234 + 2He^4.

Question: Section (B) 3: A radioactive source emits gamma rays. An experiment shows that gamma rays are emitted randomly in all directions. (a) State the random nature of radioactive decay. (b) State what gamma rays are. (i) State the safety measures that must be taken. (ii) State how the experiment should be done.

Answer: (a) Decay is random — it cannot be predicted which nucleus will break up when; only probabilities can be stated.

(b) Gamma rays are high-energy electromagnetic waves/photons from the nucleus — no charge, no rest mass. (i) Distance, short time, lead shielding, dosimeter, tongs, protective clothing. (ii) Place the source in the middle, put a detector (GM tube) at different angles and count — nearly the same count is obtained in all directions.

Question: Section (B) 4: The nuclide symbols of two isotopes of carbon are 6C^12 and 6C^14; carbon-14 changes into stable nitrogen by emitting a beta particle. (a) Which nuclide of carbon is (1) the stable isotope (2) the radioactive isotope? (b) What is meant by a beta particle? (c) Write the nuclear equation of carbon-14 changing into nitrogen by beta decay.

Answer: (a) (1) Stable isotope = 6C^12.

(2) Radioactive isotope = 6C^14.

(b) A beta particle is a fast electron emitted from the nucleus when a neutron changes into a proton.

(c) 6C^14 → 7N^14 + -1e^0 (beta particle).

Question: Section (B) 5: When a slow-moving neutron collides with the nucleus of 92U^235 uranium, it splits into two parts; in this process three neutrons and energy are also released. (a) Name the nuclear process. (b) Which isotope of U-235 and U-236 has a shorter half-life? (c) How can we prepare a radioactive isotope artificially? Give a suitable example. (d) State one use of the given process in daily life.

Answer: (a) Nuclear fission.

(b) U-236 — the compound nucleus formed is very unstable and breaks up quickly.

(c) By bombarding a stable nucleus with neutrons/particles — e.g. bombarding cobalt-59 with neutrons gives cobalt-60 (used in cancer treatment).

(d) Generating electricity in nuclear power plants.

Question: Section (B) 6: The reactions happening inside the Sun are: 2 1P^1 + 2 0n^1 → 2He^4 + energy. (a) Name the type of this reaction. (b) State the whole reaction. (c) Present a suitable argument for considering the energy produced in a nuclear fusion reaction, compared to the process of a nuclear fission reaction, as a sustainable energy source.

Answer: (a) Nuclear fusion.

(b) Light nuclei (2 protons + 2 neutrons) join to form a heavier nucleus

(helium-4) and energy is released.

(c) Fusion fuel (hydrogen) is very abundant, almost no long-lived radioactive waste is produced, and there is no risk of an uncontrolled chain reaction — so it is a safer and cleaner sustainable source.

Question: Section (B) 7: What do you understand by the half-life of a radioactive element?

Answer: Half-life is the time in which half the atoms of a radioactive sample decay — e.g. from 100 mg, 50 mg remains.

Question: Section (B) 8: (a) When radium-226 emits alpha particles it changes into radon-222; its equation is given: 88Ra^226 → 86Rn^222 + 2He^4 + energy. (b) In the given equation, consider the parent and daughter nuclides: 1. Parent element 2. Daughter element. (c) Explain the terms parent nuclide and daughter nuclide.

Answer: (b) 1.

Parent element = 88Ra^226 (radium).

2.

Daughter element = 86Rn^222 (radon).

(c) The parent nuclide is the original radioactive nuclide that decays; the daughter nuclide is the new nuclide formed after decay.

Question: Section (B) 9: (a) What are the common dangers of radioactive rays? (b) If a source emitting alpha particles enters the body, why is it more dangerous? (c) Which type of radiation is more dangerous among the types? Explain why. (d) State in detail what safety measures can be

taken to avoid the dangers of radioactive rays.

Answer: (a) Damage to cells/tissues, burns, cancer, genetic mutation.

(b) Alpha particles have the highest ionizing power; inside the body they directly damage the nearby tissues.

(c) From outside, gamma rays are the most dangerous (they penetrate the most); inside the body, alpha particles are the most dangerous.

(d) Time (short), distance (far), shielding (lead/concrete), wearing a dosimeter, holding with tongs, protective clothing, proper storage and disposal of the source.

Question: Section (C) 1: The C-14 activity in a living plant is 15.3 counts per minute per gram. In an old sample it is 3.2 counts per minute per gram. (The half-life of carbon is 5730 years.) What is the age of the sample?

Answer: Ratio = 3.2 / 15.3 = 0.21.

From (1/2)^n = 0.21, n = 2.25 half-lives.

Age = 2.25 x 5730 = 12900 years (approx).

Question: Section (C) 2: The smallest activity of carbon-14 that can be measured is about 20% of the original. If carbon-14 is used for dating a material, up to how many years can it be dated?

Answer: From (1/2)^n = 0.20, n = 2.32 half-lives.

Time = 2.32 x 5730 = 13300 years (approx).

[Note: the printed answer "1374 years" does not follow from this working — it appears misprinted.]

Question: Section (C) 3: How much time is required for 25% of the C-14 atoms in a quantity of carbon-14 to decay? (Half-life 5730 years.)

Answer: After 25% decay, 75% remain.

From (1/2)^n = 0.75, n = 0.415 half-lives.

Time = 0.415 x 5730 = 2378 years.

Question: Section (C) 4: The decay rate of carbon-14 from a sample of a healthy tree is 0.296 counts per second per gram. The decay rate of a sample found in an excavation of ancient remains is 0.109 counts per second per gram. What is the age of that sample? (Half-life 5730 years.)

Answer: Ratio = 0.109 / 0.296 = 0.368.

From (1/2)^n = 0.368, n = 1.44 half-lives.

Age = 1.44 x 5730 = 8258 years (approx).